

Aggregate Functions (Deep Notes)

What are Aggregate Functions?

Aggregate functions:

- Take **multiple input values**
- Return **one output value**

Example:

- Average marks of students
 - Total salary of employees
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List of Main Aggregate Functions

Function	Purpose
COUNT()	Counts rows
SUM()	Calculates total
AVG()	Calculates average
MIN()	Finds minimum value
MAX()	Finds maximum value

Sample Table: student

id	name	marks	city
1	Rahul	80	Nagpur
2	Priya	70	Pune
3	Aman	90	Nagpur
4	Riya	60	Pune

1 COUNT() Function

Definition

Counts number of rows.

Syntax

```
SELECT COUNT(column_name)
FROM table;
```

Examples

Count total students

```
SELECT COUNT(*) FROM student;
```

Result → 4

Count non-null values

```
SELECT COUNT(marks) FROM student;
```

Count distinct cities

```
SELECT COUNT(DISTINCT city)
FROM student;
```

2 SUM() Function

Definition

Calculates **total sum of numeric column**.

Syntax

```
SELECT SUM(column_name)
FROM table;
```

Example

```
SELECT SUM(marks)
FROM student;
```

Result → 300

3 AVG() Function

Definition

Calculates **average value**.

Syntax

```
SELECT AVG(column_name)
FROM table;
```

Example

```
SELECT AVG(marks)
FROM student;
```

Result → 75

4 MIN() Function

Definition

Finds **smallest value**.

Example

```
SELECT MIN(marks)
FROM student;
```

Result → 60

5 MAX() Function

Definition

Finds **largest value**.

Example

```
SELECT MAX(marks)
FROM student;
```

Result → 90

Aggregate Functions with GROUP BY

Aggregate functions are mostly used with **GROUP BY**.

Example: Count students per city

```
SELECT city, COUNT(*)
FROM student
GROUP BY city;
```

Result

city	count
Nagpur	2

Example: Average marks per city

```
SELECT city, AVG(marks)
FROM student
GROUP BY city;
```

Example: Total marks per city

```
SELECT city, SUM(marks)
FROM student
GROUP BY city;
```

Aggregate Functions with HAVING

Use **HAVING** to filter aggregated data.

Example: Cities with more than 1 student

```
SELECT city, COUNT(*)
FROM student
GROUP BY city
HAVING COUNT(*) > 1;
```

Example: Cities with average marks > 70

```
SELECT city, AVG(marks)
FROM student
GROUP BY city
HAVING AVG(marks) > 70;
```

Important Rules

1. Aggregate functions ignore NULL values

Example:

marks

80

NULL

70

$AVG = (80 + 70) / 2 = 75$

2. Cannot use directly in WHERE

 Wrong

```
SELECT *  
FROM student  
WHERE AVG(marks) > 70;
```

 Correct

```
SELECT city, AVG(marks)  
FROM student  
GROUP BY city  
HAVING AVG(marks) > 70;
```

3. Use with GROUP BY for meaningful results

Without GROUP BY → single result

With GROUP BY → multiple grouped results

Real-Life Example

Scenario: College Report

```
SELECT course, AVG(marks), MAX(marks), MIN(marks)  
FROM student  
GROUP BY course;
```

This gives:

- Average marks per course
- Highest marks per course
- Lowest marks per course

Combined Example

```
SELECT city,  
COUNT(*) AS total_students,  
AVG(marks) AS avg_marks,  
MAX(marks) AS highest,  
MIN(marks) AS lowest  
FROM student  
GROUP BY city;
```

Quick Summary

Function	Use
COUNT()	Number of records
SUM()	Total
AVG()	Average
MIN()	Smallest value
MAX()	Largest value